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The Editors
Physical Review D

Dear Editors,

I am pleased to submit for your consideration the manuscript *"Fixing the Dark-Energy Equation of State with Two Dimensionless Constants, φ and π: A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck 2018."*

Physics claim. The DESI results have made it pressing to ask whether the dark-energy equation of state is a set of free parameters to be measured or a structure fixed in advance. The manuscript presents a minimal-parameter framework in which the pair (w_0, w_a) follows as bare algebraic ratios of two dimensionless constants, $w_0 = -0.839950$ and $w_a = -0.669975$, with no fitted amplitude. No quantity is fitted to place it, and it sits at 0.24σ from the DESI DR2 w_0w_aCDM posterior (Pantheon+; 0.2–1.8σ across supernova compilations). It carries a pre-committed falsification criterion: exclusion at $> 3\sigma$ by DESI DR3 refutes the model, as would LiteBIRD failing to find $r = \varphi^{-10} = 8.130619 \times 10^{-3}$.

One fit, then fixed. In CMB-likelihood terms the framework is a $k = 2$ model: the background (ω_b, ω_c, n_s, w_0, w_a and the global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1}) is fixed algebraically, and only the primordial amplitude A_s and the optical depth τ are fitted, once, to Planck 2018. The full `plik` likelihood gives a best-fit χ^2 indistinguishable from ΛCDM ($\Delta\chi^2 = -2.78$) and $\Delta{\rm BIC} = -33.83$ ($k = 2$ versus $k = 6$): the preference comes from parsimony, not from a better fit. After that single fit nothing cosmological is adjusted again. Every other probe is read with the background of both models fixed in advance—SSEE at its algebraic point, ΛCDM at the Planck best fit—and with no cosmological parameter free. In every probe but one the two models tie: SSEE stays within $|\Delta\chi^2| \leq 3.0$ of ΛCDM in 10 of 11 comparisons (supernovae, CMB lensing, DES Y3, Planck, KiDS-Legacy), and BOSS full-shape gives $\Delta\chi^2 = +2.84$ against ΛCDM with its cosmology free—while ΛCDM fits up to 6 parameters and SSEE at most $\{A_s, \tau\}$, in the CMB alone. The one probe that separates them is DESI DR2 BAO, with $\chi^2 = 11.41$ for SSEE and 31.98 for ΛCDM over 13 points.

Weak lensing without an extra component. An earlier version of this work introduced an additional dark-matter component to relieve an apparent S_8 tension. Measured against the raw KiDS-Legacy cosmic-shear data (Wright et al. 2025, arXiv:2503.19441), that tension does not exist: with the algebraic background, the shear asks for $\log(10^{10}A_s) = 3.0544 \pm 0.0383$, at -0.25σ from the 3.04483 fixed by the model's own CMB fit. Fixing A_s to that value costs almost nothing ($\chi^2 = 415.59$ against 414.99 with A_s free, all remaining parameters nuisance), so $S_8 = 0.8273$ is computed rather than fitted, against 0.8265 ± 0.0176 measured. The additional component was therefore withdrawn, and a single A_s now serves the CMB and the growth of structure. KiDS-Legacy predates this analysis; I claim consistency across epochs, not priority.

Limits stated in the manuscript. The absolute scale enters as a single measured input, the local

expansion rate; the algebraic origin of ω_b and ω_c and the mechanism behind the screening ratio are open problems, listed as such; the non-linear closure of the growth amplitude is deferred to N-body work.

Originality and status. The manuscript is original, has not been published elsewhere, and is not under consideration at any other journal. A preprint of the associated framework is archived on Zenodo (concept DOI 10.5281/zenodo.19654118); it has not been posted to arXiv. I am an independent researcher without institutional affiliation and have no competing interests. All code, data provenance and validation scripts are public (https://github.com/mikealmeida1721/SSEE-Cosmologia), as stated in the manuscript's Data and Code Availability section.

Thank you for considering this submission. I would be glad to answer any questions from the editors or referees.

Sincerely,

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